Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
D		**2617170**

Enjoy a Cozy and Green Bath

Summary

A traditional bathtub cannot be reheated by itself, so users have to add hot water from time to time. Our goal is to establish a model of the temperature of bath water in space and time. Then we are expected to propose an optimal strategy for users to keep the temperature even and close to initial temperature and decrease water consumption.

To simplify modeling process, we firstly assume there is no person in the bathtub. We regard the whole bathtub as a thermodynamic system and introduce heat transfer formulas.

We establish two sub-models: adding water constantly and discontinuously. As for the former sub-model, we define the mean temperature of bath water. Introducing Newton cooling formula, we determine the heat transfer capacity. After deriving the value of parameters, we deduce formulas to derive results and simulate the change of temperature field via CFD. As for the second sub-model, we define an iteration consisting of two process: heating and standby. According to energy conservation law, we obtain the relationship of time and total heat dissipating capacity. Then we determine the mass flow and the time of adding hot water. We also use CFD to simulate the temperature field in second sub-model.

In consideration of evaporation, we correct the results of sub-models referring to some scientists' studies. We define two evaluation criteria and compare the two sub-models. Adding water constantly is found to keep the temperature of bath water even and avoid wasting too much water, so it is recommended by us.

Then we determine the influence of some factors: radiation heat transfer, the shape and volume of the tub, the shape/volume/temperature/motions of the person, the bubbles made from bubble bath additives. We focus on the influence of those factors to heat transfer and then conduct sensitivity analysis. The results indicate smaller bathtub with less surface area, lighter personal mass, less motions and more bubbles will decrease heat transfer and save water.

Based on our model analysis and conclusions, we propose the optimal strategy for the user in a bathtub and explain the reason of uneven temperature throughout the bathtub. In addition, we make improvement for applying our model in real life.

Keywords: Heat transfer, Thermodynamic system, CFD, Energy conservation

Contents

1 Introduction

1.1 Problem Background

Professional sports represent a unique intersection of athletic competition and business enterprise.While team success is traditionally measured by wins and championships, franchise owners ultimately seek to maximize long-term profit and franchise value. Player compensation, roster construction, and strategic investment decisions must therefore balance athletic contribution with economic return. These decisions are further constrained by league regulations such as salary caps, roster limits, and revenue-sharing policies, which limit purely performance-driven or profit-driven strategies.

The WNBA provides a representative setting for studying this trade-off due to its recent growth in media exposure, fan engagement, and franchise valuation alongside continued structural and financial constraints. For teams such as the Las Vegas Aces, management must evaluate players based not only on statistical performance but also on market influence, risk exposure, and future potential. This environment necessitates the development of an integrated modeling framework that quantitatively links player performance, financial outcomes, and strategic decision-making under dynamic and uncertain conditions.

1.2 Problem Restatement

To address the objectives of this problem, we must construct an integrated analytical framework that supports strategic decision-making for a professional WNBA team. Specifically, we aim to:

- **Quantifying individual player value**. We will quantify their value by incorporating both on-court performance and commercial impact.
- **Developing a financial model**. The model will link team composition, operating costs, and revenue generation to overall profitability and franchise value.
- **Establishing a dynamic decision-making mechanism**. We'll adjust roster and investment strategies in response to changes in performance, injury status, and economic conditions.
- **Evaluating player acquisition strategies**. We'll discuss strategies under league constraints such as salary caps and roster limits.
- **Assessing the impact of external structural changes**.We'll put typical changes into consideration,such as league expansion, on team profitability and competitive positioning.

1.3 Problem Analysis

The problem contains several interconnected components that require structured modeling.

First, player evaluation must extend beyond traditional performance statistics. Athletic contribution influences team wins, while popularity and market appeal affect revenue streams such as ticket sales, merchandise, and media exposure. Therefore, player value should be modeled as a multi-dimensional construct combining competitive and commercial factors.

Second, team profitability depends on both revenue generation and cost control. Revenue is influenced by team performance, player marketability, and external market conditions, whereas costs are constrained by salary structures and league regulations. A financial optimization framework is required to determine sustainable investment levels.

Third, decision-making is dynamic rather than static. Player injuries, aging effects, and economic fluctuations introduce uncertainty over time. Consequently, the model must incorporate risk and allow for strategic adjustment across multiple periods.

Finally, league-level policies, including salary caps and potential expansion, impose structural constraints that alter optimal strategies. These constraints must be explicitly embedded within the modeling framework.

2 Assumptions and Justifications

To simplify the problem and make it convenient for us to simulate real-life conditions, we make the following basic assumptions, each of which is properly justified.

- **Player performance can be represented by aggregated statistical indicators.** We assume that a player's athletic contribution can be sufficiently captured by key performance metrics such as scoring efficiency, rebounds, assists, and defensive impact, which are publicly available and measurable.
- **Commercial impact of a player can be approximated using measurable proxies.** We assume that a player's commercial value can be estimated using observable indicators such as media exposure, popularity, and merchandise influence.
- **Team revenue is positively correlated with competitive performance and player marketability.** We assume that improved team performance increases revenue streams such as ticket sales, sponsorship value, and media exposure. Additionally, high-profile players contribute independently to revenue generation.
- **Salary cap and roster constraints are binding and strictly enforced.**. We assume that league regulations, including salary caps and roster limits, restrict feasible roster construction strategies.
- **Player aging and injury risk affect future performance probabilistically.**. This hypothesis can be easily achieved in reality and will simplify our process of solving the problem.

3 Notations

Symbols	Description	Unit
h	Convection heat transfer coefficient	W/(m$^2\cdot$ K)
k	Thermal conductivity	W/(m $\cdot$ K)
c_p	Specific heat	J/(kg $\cdot$ K)
ρ	Density	kg/m^2
δ	Thickness	m
t	Temperature	°C, K
τ	Time	s, min, h
q_m	Mass flow	kg/s
Φ	Heat transfer power	W
T	A period of time	s, min, h
V	Volume	m^3, L
M, m	Mass	kg
A	Aera	m^2
a, b, c	The size of a bathtub	m^3

where we define the main parameters while specific value of those parameters will be given later.

4 Model Overview

In our basic model, we aim at three goals: keeping the temperature as even as possible, making it close to the initial temperature and decreasing the water consumption.

We start with the simple sub-model where hot water is added constantly. At first we introduce convection heat transfer control equations in rectangular coordinate system. Then we define the mean temperature of bath water.

Afterwards, we introduce Newton cooling formula to determine heat transfer capacity. After deriving the value of parameters, we get calculating results via formula deduction and simulating results via CFD.

Secondly, we present the complicated sub-model in which hot water is added discontinuously. We define an iteration consisting of two process: heating and standby. As for heating process, we derive control equations and boundary conditions. As for standby process, considering energy conservation law, we deduce the relationship of total heat dissipating capacity and time.

Then we determine the time and amount of added hot water. After deriving the value of parameters, we get calculating results via formula deduction and simulating results via CFD.

At last, we define two criteria to evaluate those two ways of adding hot water. Then we propose optimal strategy for the user in a bathtub. The whole modeling process can be shown as follows.

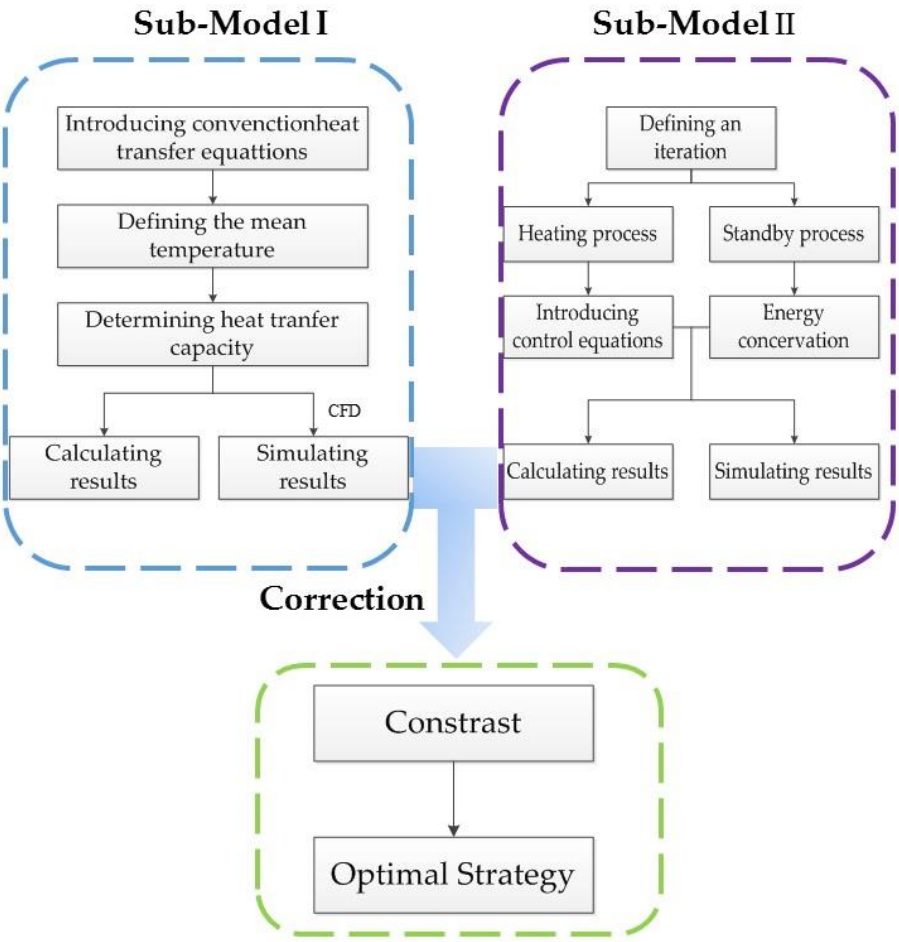

Figure 1: Modeling process

5 Sub-model I : Adding Water Continuously

We first establish the sub-model based on the condition that a person add water continuously to reheat the bathing water. Then we use Computational Fluid Dynamics (CFD) to simulate the change of water temperature in the bathtub. At last, we evaluate the model with the criteria which have been defined before.

5.1 Model Establishment

Since we try to keep the temperature of the hot water in bathtub to be even, we have to derive the amount of inflow water and the energy dissipated by the hot water into the air.

We derive the basic convection heat transfer control equations based on the former scientists' achievement. Then, we define the mean temperature of bath water. Afterwards, we determine two types of heat transfer: the boundary heat transfer and the evaporation heat transfer. Combining thermodynamic formulas, we derive calculating results. Via Fluent software, we get simulation results.

5.1.1 Control Equations and Boundary Conditions

According to thermodynamics knowledge, we recall on basic convection heat transfer control equations in rectangular coordinate system. Those equations show the relationship of the temperature of the bathtub water in space.

We assume the hot water in the bathtub as a cube. Then we put it into a rectangular coordinate system. The length, width, and height of it is a, b and c.

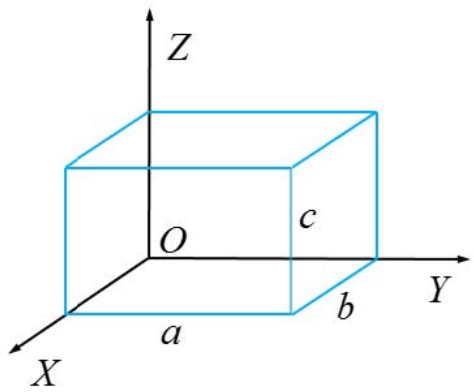

Figure 2: Modeling process

In the basis of this, we introduce the following equations [**5**]:

– **Continuity equation:**

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0 \tag{1}$$

where the first component is the change of fluid mass along the X-ray. The second component is the change of fluid mass along the Y-ray. And the third component is the change of fluid mass along the Z-ray. The sum of the change in mass along those three directions is zero.

– **Moment differential equation (N-S equations):**

$$\begin{cases} \rho\Big(u\dfrac{\partial u}{\partial x}+v\dfrac{\partial u}{\partial y}+w\dfrac{\partial u}{\partial z}\Big)=-\dfrac{\partial p}{\partial x}+\eta\Big(\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}+\dfrac{\partial^2 u}{\partial z^2}\Big) \\ \rho\Big(u\dfrac{\partial v}{\partial x}+v\dfrac{\partial v}{\partial y}+w\dfrac{\partial v}{\partial z}\Big)=-\dfrac{\partial p}{\partial y}+\eta\Big(\dfrac{\partial^2 v}{\partial x^2}+\dfrac{\partial^2 v}{\partial y^2}+\dfrac{\partial^2 v}{\partial z^2}\Big) \\ \rho\Big(u\dfrac{\partial w}{\partial x}+v\dfrac{\partial w}{\partial y}+w\dfrac{\partial w}{\partial z}\Big)=-g-\dfrac{\partial p}{\partial z}+\eta\Big(\dfrac{\partial^2 w}{\partial x^2}+\dfrac{\partial^2 w}{\partial y^2}+\dfrac{\partial^2 w}{\partial z^2}\Big) \end{cases} \tag{2}$$

– **Energy differential equation:**

$$\rho c_p\Big(u\frac{\partial t}{\partial x}+v\frac{\partial t}{\partial y}+w\frac{\partial t}{\partial z}\Big)=\lambda\Big(\frac{\partial^2 t}{\partial x^2}+\frac{\partial^2 t}{\partial y^2}+\frac{\partial^2 t}{\partial z^2}\Big) \tag{3}$$

where the left three components are convection terms while the right three components are conduction terms.

By Equation (**??**), we have .....

.....

On the right surface in Fig. **??**, the water also transfers heat firstly with bathtub inner surfaces and then the heat comes into air. The boundary condition here is .....

5.1.2 Definition of the Mean Temperature

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5.1.3 Determination of Heat Transfer Capacity

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6 Sub-model II: Adding Water Discontinuously

In order to establish the unsteady sub-model, we recall on the working principle of air conditioners. The heating performance of air conditions consist of two processes: heating and standby. After the user set a temperature, the air conditioner will begin to heat until the expected temperature is reached. Then it will go standby. When the temperature get below the expected temperature, the air conditioner begin to work again. As it works in this circle, the temperature remains the expected one.

Inspired by this, we divide the bathtub working into two processes: adding hot water until the expected temperature is reached, then keeping this condition for a while unless the temperature is lower than a specific value. Iterating this circle ceaselessly will ensure the temperature kept relatively stable.

6.1 Heating Model

6.1.1 Control Equations and Boundary Conditions

6.1.2 Determination of Inflow Time and Amount

6.2 Standby Model

6.3 Results

We first give the value of parameters based on others' studies. Then we get the calculation results and simulating results via those data.

6.3.1 Determination of Parameters

After establishing the model, we have to determine the value of some important parameters.

As scholar Beum Kim points out, the optimal temperature for bath is between 41 and 45°C [1]. Meanwhile, according to Shimodozono's study, 41°C warm water bath is the perfect choice for individual health [2]. So it is reasonable for us to focus on $41°\text{C} \sim 45°\text{C}$. Because adding hot water continuously is a steady process, so the mean temperature of bath water is supposed to be constant. We value the temperature of inflow and outflow water with the maximum and minimum temperature respectively.

The values of all parameters needed are shown as follows:

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Table 1: The calculating results

Variables	Values	Unit
A_1	1.05	m^2
A_2	2.24	m^2
Φ_1	189.00	W
Φ_2	43.47	W
Φ	232.47	W
q_m	0.014	g/s

6.3.2 Calculating Results

Putting the above value of parameters into the equations we derived before, we can get the some data as follows:

From Table **??**, ...

.....

7 Correction and Contrast of Sub-Models

After establishing two basic sub-models, we have to correct them in consideration of evaporation heat transfer. Then we define two evaluation criteria to compare the two sub-models in order to determine the optimal bath strategy.

7.1 Correction with Evaporation Heat Transfer

Someone may confuse about the above results: why the mass flow in the first sub-model is so small? Why the standby time is so long? Actually, the above two sub-models are based on ideal conditions without consideration of the change of boundary conditions, the motions made by the person in bathtub and the evaporation of bath water, etc. The influence of personal motions will be discussed later. Here we introducing the evaporation of bath water to correct sub-models.

7.2 Contrast of Two Sub-Models

Firstly we define two evaluation criteria. Then we contrast the two submodels via these two criteria. Thus we can derive the best strategy for the person in the bathtub to adopt.

8 Model Analysis and Sensitivity Analysis

8.1 The Influence of Different Bathtubs

Definitely, the difference in shape and volume of the tub affects the convection heat transfer. Examining the relationship between them can help people choose optimal bathtubs.

8.1.1 Different Volumes of Bathtubs

In reality, a cup of water will be cooled down rapidly. However, it takes quite long time for a bucket of water to become cool. That is because their volume is different and the specific heat of water is very large. So that the decrease of temperature is not obvious if the volume of water is huge. That also explains why it takes 45 min for 320 L water to be cooled by 1°C.

In order to examine the influence of volume, we analyze our sub-models by conducting sensitivity Analysis to them.

We assume the initial volume to be 280 L and change it by $\pm 5\%$, $\pm 8\%$, $\pm 12\%$ and $\pm 15\%$. With the aid of sub-models we established before, the variation of some parameters turns out to be as follows

Table 2: Variation of some parameters

V	A_1	A_2	T_2	q_{m1}	q_{m2}	Φ_q
-15.00%	-5.06%	-9.31%	-12.67%	-2.67%	-14.14%	-5.80%
-12.00%	-4.04%	-7.43%	-10.09%	-2.13%	-11.31%	-4.63%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%

9 Strength and Weakness

9.1 Strength

- We analyze the problem based on thermodynamic formulas and laws, so that the model we established is of great validity.
- Our model is fairly robust due to our careful corrections in consideration of real-life situations and detailed sensitivity analysis.
- Via Fluent software, we simulate the time field of different areas throughout the bathtub. The outcome is vivid for us to understand the changing process.
- We come up with various criteria to compare different situations, like water consumption and the time of adding hot water. Hence an overall comparison can be made according to these criteria.
- Besides common factors, we still consider other factors, such as evaporation and radiation heat transfer. The evaporation turns out to be the main reason of heat loss, which corresponds with other scientist's experimental outcome.

9.2 Weakness

- Having knowing the range of some parameters from others' essays, we choose a value from them to apply in our model. Those values may not be reasonable in reality.
- Although we investigate a lot in the influence of personal motions, they are so complicated that need to be studied further.
- Limited to time, we do not conduct sensitivity analysis for the influence of personal surface area.

10 Further Discussion

In this part, we will focus on different distribution of inflow faucets. Then we discuss about the real-life application of our model.

- Different Distribution of Inflow Faucets

  In our before discussion, we assume there being just one entrance of inflow.

  From the simulating outcome, we find the temperature of bath water is hardly even. So we come up with the idea of adding more entrances.

  The simulation turns out to be as follows

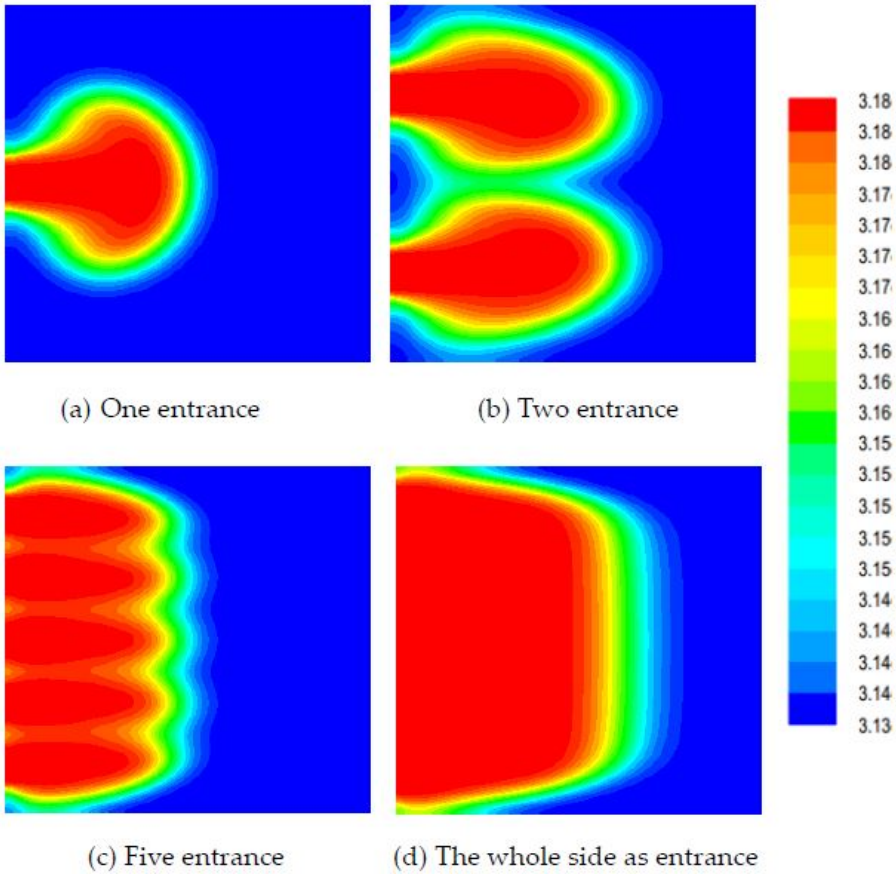

Figure 3: The simulation results of different ways of arranging entrances

  From the above figure, the more the entrances are, the evener the temperature will be. Recalling on the before simulation outcome, when there is only one entrance for inflow, the temperature of corners is quietly lower than the middle area.

  In conclusion, if we design more entrances, it will be easier to realize the goal to keep temperature even throughout the bathtub.

- Model Application

  Our before discussion is based on ideal assumptions. In reality, we have to make some corrections and improvement.

  1) Adding hot water continually with the mass flow of 0.16 kg/s. This way can ensure even mean temperature throughout the bathtub and waste less water.
  2) The manufacturers can design an intelligent control system to monitor the temperature so that users can get more enjoyable bath experience.

3) We recommend users to add bubble additives to slow down the water being cooler and help cleanse. The additives with lower thermal conductivity are optimal.
4) The study method of our establishing model can be applied in other area relative to convection heat transfer, such as air conditioners.

References

[1] Gi-Beum Kim. Change of the Warm Water Temperature for the Development of Smart Healthecare Bathing System. Hwahak konghak. 2006, 44(3): 270-276.

[2] https://en.wikipedia.org/wiki/Convective_heat_transfer#Newton.27s_law_of_cooling

[3] https://en.wikipedia.org/wiki/Navier%E2%80%93Stokes_equations

[4] https://en.wikipedia.org/wiki/Computational_fluid_dynamics

[5] Holman J P. Heat Transfer (9th ed.), New York: McGraw-Hill, 2002.

[6] Liu Weiguo, Chen Zhaoping, ZhangYin. Matlab program design and application. Beijing: Higher education press, 2002. (In Chinese)

Dear XXX:

From simulation results of real-life situations, we find it takes a period of time for the inflow hot water to spread throughout the bathtub. During this process, the bath water continues transferring heat into air, bathtub and the person in bathtub. The difference between heat transfer capacity makes the temperature of various areas to be different. So that it is difficult to get an evenly maintained temperature throughout the bath water.

In order to enjoy a comfortable bath with even temperature of bath water and without wasting too much water, we propose the following suggestions.

- Adding hot water consistently
- Using smaller bathtub if possible
- Decreasing motions during bath
- Using bubble bath additives
- Arranging more faucets of inflow

Sincerely yours,

Your friends

Appendices

Appendix A First appendix

In addition, your report must include a letter to the Chief Financial Officer (CFO) of the Goodgrant Foundation, Mr. Alpha Chiang, that describes the optimal investment strategy, your modeling approach and major results, and a brief discussion of your proposed concept of a return-on-investment (ROI). This letter should be no more than two pages in length.

Here are simulation programmes we used in our model as follow.

Input matlab source:

```
function [t,seat,aisle]=OI6Sim(n,target,seated)
pab=rand(1,n);
for i=1:n
    if pab(i)<0.4
       aisleTime(i)=0;
    else
        aisleTime(i)=trirnd(3.2,7.1,38.7);
    end
end
```

Appendix B Second appendix

some more text **Input C++ source:**

```
//
    ==============================================================================
// Name        : Sudoku.cpp
// Author      : wzlf11
// Version     : a.0
// Copyright   : Your copyright notice
// Description : Sudoku in C++.
//
    ==============================================================================

#include <iostream>
#include <cstdlib>
#include <ctime>

using namespace std;

int table[9][9];

int main() {

    for(int i = 0; i < 9; i++){
        table[0][i] = i + 1;
    }

    srand((unsigned int)time(NULL));
```

```

    shuffle((int *)&table[0], 9);

    while(!put_line(1))
    {
        shuffle((int *)&table[0], 9);
    }

    for(int x = 0; x < 9; x++){
        for(int y = 0; y < 9; y++){
            cout << table[x][y] << " ";
        }

        cout << endl;
    }

    return 0;
}
```

Report on Use of AI

1. OpenAI ChatGPT (Nov 5, 2023 version, ChatGPT-4,)

   Query1: <insert the exact wording you input into the AI tool>
   Output: <insert the complete output from the AI tool>

2. OpenAI Ernie (Nov 5, 2023 version, Ernie 4.0)

   Query1: <insert the exact wording of any subsequent input into the AI tool>
   Output: <insert the complete output from the second query>

3. Github CoPilot (Feb 3, 2024 version)

   Query1: <insert the exact wording you input into the AI tool>
   Output: <insert the complete output from the AI tool>

4. Google Bard (Feb 2, 2024 version)

   Query1: <insert the exact wording of your query>
   Output: <insert the complete output from the AI tool>